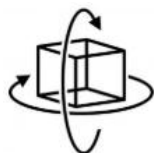


AHRS IMU Sensor | WTGAHRS3

The Robust Acceleration, Angular velocity, Angle , Magnetic filed & Air Pressure & GPS Monitor Detector

The WTGAHRS3 is a IMU sensor device, detecting acceleration, angular velocity, angle , magnetic filed , air pressure as well as gps data. The robust housing and the small outline makes it perfectly suitable for industrial applications such as condition monitoring and predictive maintenance. Configuring the device enables the customer to address a broad variety of application by interpreting the sensor data by smart algorithms and Kalman filtering.

BUILT-IN SENSORS



Accelerometer



Gyroscope



Magnetometer



Barometer



GPS Module



Tutorial Link

[Google Drive](#)

Link to instructions DEMO:
[WITMOTION Youtube Channel](#)

If you have technical problems or cannot find the information that you need in the provided documents, please contact our support team. Our engineering team is committed to providing the required support necessary to ensure that you are successful with the operation of our AHRS sensors.

Contact

[Technical Support Contact Info](#)

Application

- AGV Truck
- Platform Stability
- Auto Safety System
- 3D Virtual Reality
- Industrial Control
- Robot
- Car Navigation
- UAV
- Truck-mounted Satellite Antenna Equipment

Contents

Tutorial Link	- 2 -
Contact	- 2 -
Application	- 2 -
Contents	- 3 -
1 Overview	- 4 -
2 Features	- 5 -
3 Specification	- 6 -
3.1 Parameter	- 6 -
Accelerometer Parameters	- 7 -
Gyroscope parameters	- 8 -
Pitch and roll angle parameters	- 9 -
Heading angle parameter	- 9 -
GPS parameter	- 10 -
Module parameters	- 12 -
Basic parameters	- 12 -
Electrical parameters	- 12 -
3.2 Axial Direction	- 13 -
4 Communication Protocol	- 13 -

1 Overview

WTGAHRS3's scientific name is AHRS IMU sensor. A sensor measures 3-axis angle, angular velocity, acceleration, magnetic field, air pressure and GPS data. Its strength lies in the algorithm which can calculate three-axis angle accurately.

WTGAHRS3 is employed where the highest measurement accuracy is required. WTGAHRS3 offers several advantages over competing sensor:

- Heated for best data availability: new WITMOTION patented zero-bias automatic detection calibration algorithm outperforms traditional accelerometer sensor
- High precision Roll Pitch Yaw (X Y Z axis) Acceleration + Angular Velocity + Angle + Magnetic Field + Air Pressure + GPS data output
- Low cost of ownership: remote diagnostics and lifetime technical support by WITMOTION service team
- Developed tutorial: providing manual, datasheet, Demo video, PC software, mobile phone APP, and 51 serial, STM32, Arduino, and Matlab sample code, communication protocol
- WITMOTION sensors have been praised by thousands of engineers as a recommended attitude measurement solution

2 Features

- Built-in WT901B + GPS + BeiDou module, for detailed parameters, please refer to the instructions of WT901B.
- The default baud rate of this device is 9600 and could be changed.
- The interface of this product only leads to a serial port
- The module consists of a high precision gyroscope, accelerometer, geomagnetic field and barometer sensor. The product can solve the current real-time motion posture of the module quickly by using the high-performance microprocessor, advanced dynamic solutions and Kalman filter algorithm.
- The advanced digital filtering technology of this product can effectively reduce the measurement noise and improve the measurement accuracy.
- Maximum 200Hz data output rate. Output content can be arbitrarily selected, the output speed 0.2HZ~ 200HZ adjustable.

3 Specification

3.1 Parameter

Parameter	Specification
➤ Working voltage	5 ~36V
➤ Current	<40mA (12V)
➤ Size	40mm x 32mm X 16mm 1.575' x 1.260' x0.630'
➤ Measurement dimension	Angle: X Y Z, 3-axis Acceleration: X Y Z, 3-axis Angular Velocity: X Y Z, 3-axis GPS: X Y Z, 3-axis
➤ Range	XZ-axis: $\pm 180^\circ$ Y-axis: $\pm 90^\circ$
➤ Output frequency	0.2Hz--200Hz
➤ Stability	Acceleration: 0.01g Angular Velocity: 0.05°/s.
➤ Attitude stability	0.01°
➤ Angle accuracy	Static 0.05°, Dynamic 0.1°
➤ Interface	Serial TTL level, (baud rate supports 4800, 9600, 19200, 38400, 57600, 115200, 230400)
➤ Output content	Time, acceleration, angular velocity, angle, port status, four elements, latitude and longitude, ground speed
➤ GPS	GPS+Beidou signal
➤ Applicable environment	Dynamic and static
➤ Antenna	External
➤ Inertial navigation	Built-in inertial navigation algorithm

Accelerometer Parameters

Parameter	Condition	Typical Value
Range		$\pm 16g$
Resolution	$\pm 16g$	0.0005(g/LSB)
RMS noise	Bandwidth = 100Hz	0.75~1mg-rms
Static zero drift	Placed horizontally	$\pm 80mg$
Temperature drift	$-40^{\circ}C \sim +85^{\circ}C$	$\pm 0.15mg/^{\circ}C$
Bandwidth		5~256Hz

Gyroscope parameters

Parameter	Condition	Typical Value
Range		$\pm 2000^{\circ}/s$
Resolution	$\pm 2000^{\circ}/s$	$0.061(^{\circ}/s)/(LSB)$
RMS noise	Bandwidth = 100Hz	$0.05^{\circ}/s-rms$
Static zero drift	Placed horizontally	$\pm 0.5 \sim 1^{\circ}/s$
Temperature drift	$-40^{\circ}C \sim +85^{\circ}C$	$\pm 0.02\%/^{\circ}C$
Bandwidth		$5 \sim 256Hz$

Pitch and roll angle parameters

Parameter	Condition	Typical Value
Range		X: $\pm 180^\circ$
		Y: $\pm 90^\circ$
Inclination accuracy		0.2°
Resolution	Placed horizontally	0.0055°
Temperature drift	-40°C ~ +85°C	$\pm 0.5 \sim 1^\circ$

Heading angle parameter

Parameter	Condition	Typical Value
Range		Z: $\pm 180^\circ$
Heading accuracy	6-axis algorithm, static	0.5° (Dynamic integral cumulative error exists)
Resolution		0.0055°

GPS parameter

Index	Parameter
Signal Reception	BDS/GPS/QZSS/GLONASS
Cold Start TTFF	$\leq 32s$
Hot Start TTFF	$\leq 1s$
Recapture TTFF	$\leq 1s$
Cold Start Capture Sensitivity	-148dBm
Hot Start Capture Sensitivity	-156dBm
Recapture Sensitivity	-160dBm
Tracking Sensitivity	-162dBm
Positioning Precision	<2.5m (CEP50)
Speed Measurement Accuracy	<0.1m/s (1σ)
Positioning Update Rate	5Hz
Protocol	NMEA0183

Maximum Height	18000m
Maximum Speed	515m/s
Maximum Acceleration	4g

Module parameters

Basic parameters

Parameter	Condition	Min	Default	Max
Interface	UART	4800bps	9600bps	230400bps
Output content		Time, acceleration, angular velocity, angle, latitude and longitude, ground speed		
Output rate		0.2Hz	10Hz	200Hz
Start time				1000ms
Operating temperature		-40℃		85℃
Storage temperature		-40℃		100℃
Shockproof				20000g

Electrical parameters

Parameter	Condition	Min	Default	Max
Supply voltage		5V	12V	36V
Working current	Work (12V)		16.1mA(TTL) 17.5mA	

			(232)	
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3.2 Axial Direction



As shown in the figure, the X axis is to the right, the Y axis is up, and the Z axis is the vertical module outward. The direction of rotation is defined by the right-hand rule, that is, the thumb of the right hand points to the axial direction, and the direction in which the four fingers bend is the direction of rotation around the axis. The angle of X, Y, Z axis is the angle of rotation around X, Y, Z axis

4 Communication Protocol

Level: TTL level

Baud rate: 4800, 9600 (default), 19200 38400, 57600, 115200, 230400, stop bit and parity

[Link to WITMOTION protocol.](#)

